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An auction-enabled collaborative routing mechanism for omnichannel on-demand logistics through transshipment

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ABSTRACT

Nowadays' on-demand logistics operations are carried out on separate delivery networks from competitive contractors, with redundant resources and high costs. This article proposes a new paradigm to deal with industrial and societal challenges, developing an auction-enabled collaborative routing mechanism for omnichannel on-demand logistics in a real-time transshipment network. We consider an online service platform for real-time management of on-demand pickup and delivery tasks, where multiple freight shippers can trade with multiple freight carriers. Freight shippers are retailers or individual customers, while freight carriers are a group of logistics service providers. The platform acts as the auctioneer. A two-stage combinational auction mechanism is designed for dynamic on-demand task (re)allocation. Tactical-level auctioning and operational-level routing decisions are optimized together. The transshipment-based task generation method is used to identify uneconomic paths from carrier's real-time network for outsourcing. A transshipment-based routing algorithm is developed to enable each carrier to make decentralized decisions for network reconstruction and transportation bidding. Our model aims to improve the overall social welfare while bringing benefits to stakeholders involved. The computational results have shown positive society impacts. Specifically, shippers' payments can be saved while carriers' profits are increased compared with other operative models which have been investigated in previous research studies or industry. In addition, a substantial reduction in CO_2 emissions and vehicles required can be achieved. The main reason for the improvement in social welfare is due to the optimal network achieved through collaboration. We also numerically analyze the impacts of three key factors: growth in demand density, urgency of tasks and flexible auction interval.

1. Introduction

Global retail ecommerce sales are expected to increase to US$ 6.5 trillion by the end of 2023 (eMarketer, 2019). The rapid growth of this market leads to on-demand logistics in which delivery orders from both online or offline channels are immediately fulfilled with customer desired time, place and quantities (Hong et al., 2019). Increased service levels expressed in hours to deliver, is also rigidly required. Currently, most on-demand logistics operations are carried out on separate delivery networks from different professional or crowdsourced delivery contractors, with their own dedicated facilities and fleets (Bányai et al., 2018). Many companies face

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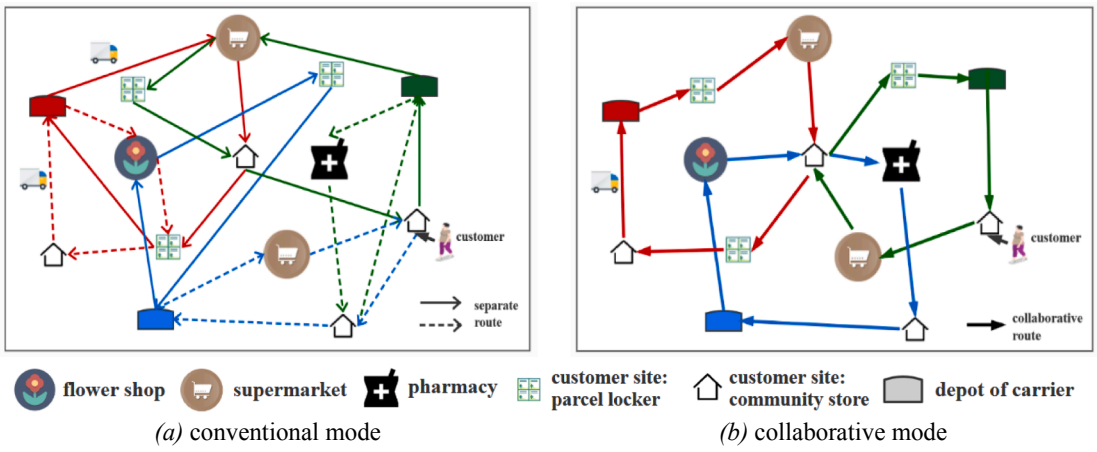


Fig. 1. Conventional mode vs. collaborative mode for on-demand logistics.

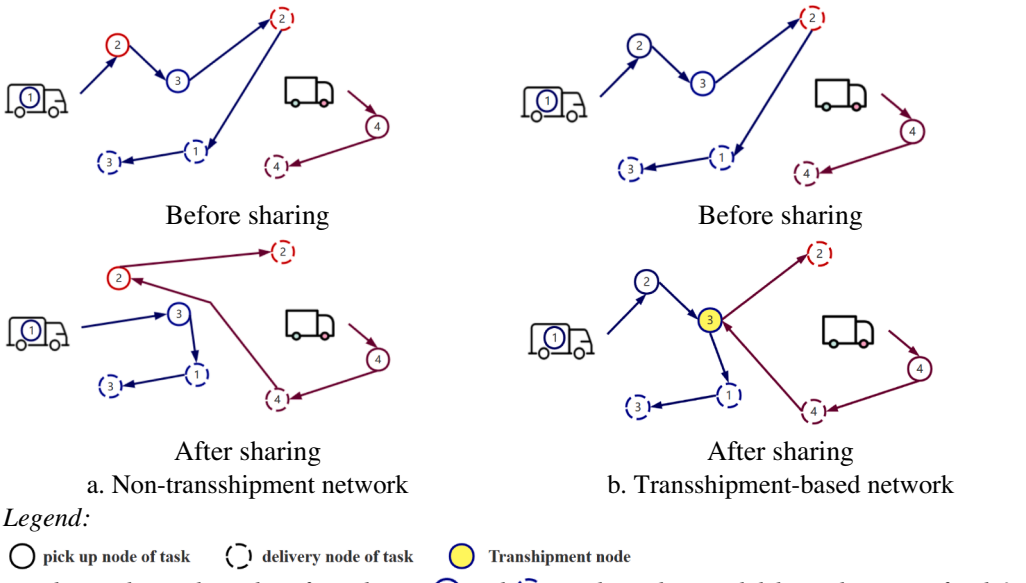


Fig. 2. Non-transshipment and transshipment-based collaborative routing.

difficulties controlling omnichannel order fulfillment and reducing on-demand delivery costs. Further, although a set of accessible pickup locations (e.g., smart lockers and community stores) exist for every customer, these depots are not employed for cargo transshipment and consolidation.

This article investigates a new paradigm defined as an auction-enabled collaborative routing mechanism with transshipment (ACRT) for omnichannel on-demand logistics through resource sharing. We consider a service platform for the real-time management of on-demand pickup and delivery tasks, where multiple task shippers can trade with multiple task carriers. Task shippers are retailers in the omnichannel ecosystem who could be any grocery, shopping mall, pharmacy or flower shop. Individual customers can also be a shipper since the on-demand delivery of consumer-to-consumer direct shipments are widely operated within cities. Task carriers refer to a group of logistics service providers who own sufficient resources such as vehicles and drivers to complete on-demand deliveries. Effective transportation planning is able to improve social welfare including all stakeholders' benefits or costs (Rouhani et al., 2016). In this article, the proposed ACRT aims to bring about positive social welfare with novel optimization mechanisms and technologies for shippers, carriers and city residents. Social welfare improvements involve three dimensions (i) shippers: payment for logistics services; (ii) carriers: transportation profit; (iii) city residents: reduction in greenhouse emissions.

Fig. 1 illustrates the differences between the conventional mode *(a)* and the transshipment-based collaborative mode *(b)* for omnichannel on-demand logistics. In the conventional delivery mode, shippers select carriers from the transportation market randomly or based on experience. Service pricing is usually fixed within a certain delivery area. Carriers process tasks separately, they

are unable to cope with lumpy demands during peak hours but become under-utilized during off-peak periods. Carriers are self-interested and independent to each other while making their own decisions. In the proposed ACRT, shippers can purchase transport services more economically from an auction-enabled on-demand service platform. Carriers can outsource tasks through an arbitrary open facility to optimize delivery plans with reduced costs. Parcel lockers or community stores near the customers' house can be utilized for transshipment.

It has been demonstrated that auction mechanisms provide a very effective way for collaboration in logistics (Karaenke et al., 2019). Several studies have addressed auction approaches in decentralized transportation optimization with carriers' collaboration. Existing models are either focused on task allocation from shippers to carriers (Andres Figliozzi et al., 2003; Figliozzi et al., 2004; Figliozzi et al., 2005; Mes et al., 2007; Mes et al., 2013) or on shared task reallocation among carriers (Berger and Bierwirth 2010; Dai and Chen 2011; Dai et al., 2014; Gansterer and Hartl 2016; Lyu et al., 2019). The problems for freight task sharing were examined in a static context where all tasks were revealed before planning. Social welfare for shippers and residents is scarcely considered. Here, we incorporate the allocation of new tasks and shared tasks reallocation through dynamic on-line auctions. To the best of our knowledge, there are limited studies considering combined decision-making for multiple stakeholders (including both shippers and carriers) with multiple objectives for on-demand logistics.

To enable more flexible trading and sharing in the carrier coalitions context, real-time transshipment is integrated into the ACRT paradigm. Dynamic tasks are allowed to be executed by multiple carriers. The advantage of transshipment-based collaborative routing is illustrated in Fig. 2. In a non-transshipment network, the paired pickup and delivery node of task 2 of one seller carrier have to be outsourced together to another actual carrier (Fig. 2a). In contrast, only the delivery node of task 2 is outsourced in the transshipment-based network (transshipped via the pickup node of task 3), instead of outsourcing the entire task 2 (Fig. 2b).

The routing component of the ACRT is a modified version of the dynamic pickup and delivery problem with transshipment and time windows (DPDPTTW). There has been a limited amount of research that has focused on the DPDPTTW in the transportation science and operations research literature. Thangiah et al. (2007) adapted the P&D heuristic of Shang and Cuff (1996) in examining DPDPTTW applications, however, the P&D heuristic in Shang and Cuff (1996) was designed to solve the static PDPTTW. In addition, the detailed algorithm for the design of transshipment insertion is missing in both studies. The above gaps constitute one of the objectives of this article.

In particular, this article aims to answer the following questions:

1. What is an effective auction mechanism for the optimal procurement of on-demand delivery services as well as carrier collaborations in the ACRT paradigm?
2. What is a dynamic task generation strategy for determining the most uneconomic path from a carrier's network?
3. How can the dynamic transshipment-based routing problem with ACRT requirements be solved?
4. What are the impacts of key factors (e.g., demand density growth, urgency of tasks and auction interval) on individual, and total logistics network performance?

To cope with the complex ACRT problem, three mechanisms are designed including, (i) a two-stage auction mechanism (TAM), (ii) a transshipment-based task generation mechanism (TTGM), and (iii) a transshipment-based routing mechanism (TRM) (see Section 4). Compared with the conventional mode, the ACRT is evaluated via simulated omnichannel scenarios like business-to-customer (B2C) fulfillment, customer-to-customer (C2C) fulfillment and hybrid fulfillment of B2C and C2C. We further analyze the impact of key factors on the performance of the ACRT to investigate the practical significance of the proposed solution (see Section 5). A list of abbreviations used in this article can be found in Appendix A.

This article is organized as follows. In Section 2, we review the relevant literature. We describe the new paradigm of the ACRT for omnichannel on-demand logistics in Section 3. In Section 4, integrated methods of the ACRT involving TAM, TTGM and TRM are proposed. We present a computational study in Section 5 to quantitatively assess the solution feasibility with associated key impact analyses. Our conclusions and future research directions are given in Section 6.

2. Literature review

The related literature can be categorized into three dimensions: (1) technology-driven on-demand logistics; (2) auction-based transportation service trading; and (3) transshipment-based dynamic pickup and delivery. Finally, research gaps are summarized.

2.1. Technology-driven on-demand logistics

On-demand logistics has become an important trend in omnichannel ecommerce as more and more retailers are promising same-day delivery (Lafkihi et al., 2019a). It has been thoroughly investigated for passenger transportation such as Uber and Didi but not for freight transportation in the first/last-mile context. Some innovative solutions have been studied to deal with issues in this area, which can be categorized into three main research streams. The first stream is collaborative freight transportation in urban areas, which was surveyed and analyzed by Cleophas et al. (2019) in terms of vertical and horizontal approaches of collaboration from an operations research perspective. This concerns the dynamic optimization of delivery routes in a collaborative setting, with or without capacity and time window constraints, see e.g. Thompson and Hassall (2012), Montoya-Torres et al. (2016), Park et al. (2016) and Gansterer and Hartl (2018). The second stream focuses on the invention of new delivery options enabled by the Internet-of-Things (IoT) and digital connectivity, such as cargo bicycles, electric autonomous vehicles (e.g., AGVs, drones), and smart parcel lockers (Zhang et al., 2018;

Slabinac, 2015; Deutsch and Golany, 2018; Faugère and Montreuil, 2018). The third stream focuses on the implementation of crowdsourced delivery (Mehmann et al., 2015), either by building a network of neighbors in the case of home delivery failure (Akeb et al., 2018), or by using taxis to collect citywide ecommerce reverse flows (Chen et al., 2017).

A hyperconnected city logistics strategy is designed by exploiting key concepts of the Physical Internet, which is highly related to the proposed on-demand logistics paradigm. The Physical Internet (PI) promotes the creation of delivery channels for innovative services, regardless of whether they are a part of data exchange or logistics. This term is included as part of the Internet of Services (Schroth and Janner, 2007). Sallez et al. (2016) showed that PI-designed containers can be advantageous. A systematic framework for designing efficient and sustainable city logistics and transportation systems enabled by the PI was presented by Crainic and Montreuil (2016). Mohamed et al. (2017) explored the operational urban transportation problem of PI-containers in hyperconnected cities, where goods in a container were optimally routed and transported. This solution can reduce total transportation costs. However, the development and implementation of the PI still faces many challenges (Giusti et al., 2019). Meyer et al. (2019) claimed that trustful collaboration and leadership concerns are major barriers for the implementation of the PI. In addition, Qiao et al. (2019) proposed a dynamic pricing model for transportation service providers using the PI.

2.2. Auction-enabled transportation service trading

Driven by booming ecommerce marketplaces, auction-based trading mechanisms in both truckload and less-than-truckload transportation industries have been thoroughly studied in the context of horizontal collaboration to optimize vehicle capacity utilization, reduce the number of redundant trips and reduce excessive costs per delivery (Xu et al., 2017; Figliozzi et al., 2005, Lafkihi et al., 2019b, Sun et al., 2019). But the growth in demand and increasingly fragmented tasks for on-demand logistics services burdens first/last-mile transportation networks with increased volume and a higher level of pickup and delivery complexity, calling for tighter coordination and consolidation of shipments (Bergmann et al., 2020; Liang et al., 2020). This article investigates auction trading to deal with this issue, which can be further classified into two types: (1) task allocation/pricing between shippers and carriers, and (2) task reallocation and sharing with carrier collaboration. For task allocation/pricing between shippers and carriers, an auction-based framework was proposed to allocation freight jobs from shippers to carriers for the truckload pickup and delivery problem in a dynamic context (Figliozzi et al., 2003). Based on this framework, a set of extended problems have been discussed with time-sensitive truckload pickup-and-delivery requests, including optimal assignment strategies using sequential auctions. Mes et al., (2009) examined real-time scheduling of pickup-and-delivery jobs through a multi-agent system where shippers and vehicle agents use profit maximizing strategies. Compared with centralized benchmarks, the system-wide costs were estimated to be reduced by 10–20%.

For task reallocation and sharing with carrier collaboration, Figliozzi (2006) addressed transport tasks reallocation among carriers. In their model, each carrier can share their single/bundle tasks dynamically once new tasks have been revealed. However, the effect of bundle size on system efficiency was not discussed. Similar multi-step single-round auction-based sharing models for task reallocation have been investigated (Berger and Bierwirth, 2010; Gansterer and Hartl, 2016; Lyu et al., 2019). Multi-round auctions have also been developed and applied in transport task sharing models (Dai and Chen, 2011; Dai et al., 2014; Wang and Kopfer, 2014).

2.3. Transshipment-based dynamic pickup and delivery

In the studied ACRT problem, the routing mechanism is a modified version of the dynamic pickup and delivery problem with transshipment and time windows (DPDPTTW). Only a limited amount of research has investigated the DPDPTTW (Baldacci et al., 2017). The most related research streams are the pickup and delivery problem with transshipment and time windows (PDPTTW) and the dynamic pickup and delivery problem (DPDP). A survey of solution methods for the PDPTTW can be found in Danloup et al. (2018). General issues and solution approaches for the DPDP are discussed by Berbeglia et al. (2010) and Arslan et al. (2019). Bouros et al., (2011) proposed a graph-based heuristic for the DPDP with transshipment without considering time windows. The performance of the heuristic was evaluated by comparing it with the Tabu-search based insertion method. Vornhusen et al. (2014) examined the PDPTTW in a collaborative context using a mixed-integer programming model. Vehicles involved in transferring the same task were synchronized at transshipment depots. However, it lacks efficient and effective heuristic approaches. Thangiah et al. (2007) investigated the design and implementation of heuristics to cope with the DPDPTTW. Their solution approach is adapted from the P&D heuristic in that was developed to solve the static PDPTTW (Shang and Cuff, 1996). The general procedure of the P&D heuristic is that when assigning a task, the first choice is to insert both the pickup and delivery nodes into a route without transshipment. If non-transshipment insertion is unattainable, the task is to be inserted with a transshipment between two routes. Constructing a mini route (i.e., a new route constructed by several unassigned tasks) is performed when neither insertion is attainable. With this heuristic, freight transshipment only happens when non-transshipment insertion fails. This may lead to a less effective result using transshipment. In addition, details of how to optimize transshipment insertion were not described in either Shang and Cuff, (1996) or Thangiah et al. (2007).

2.4. Summary

Based on the above review of the literature, we identified the following research gaps.

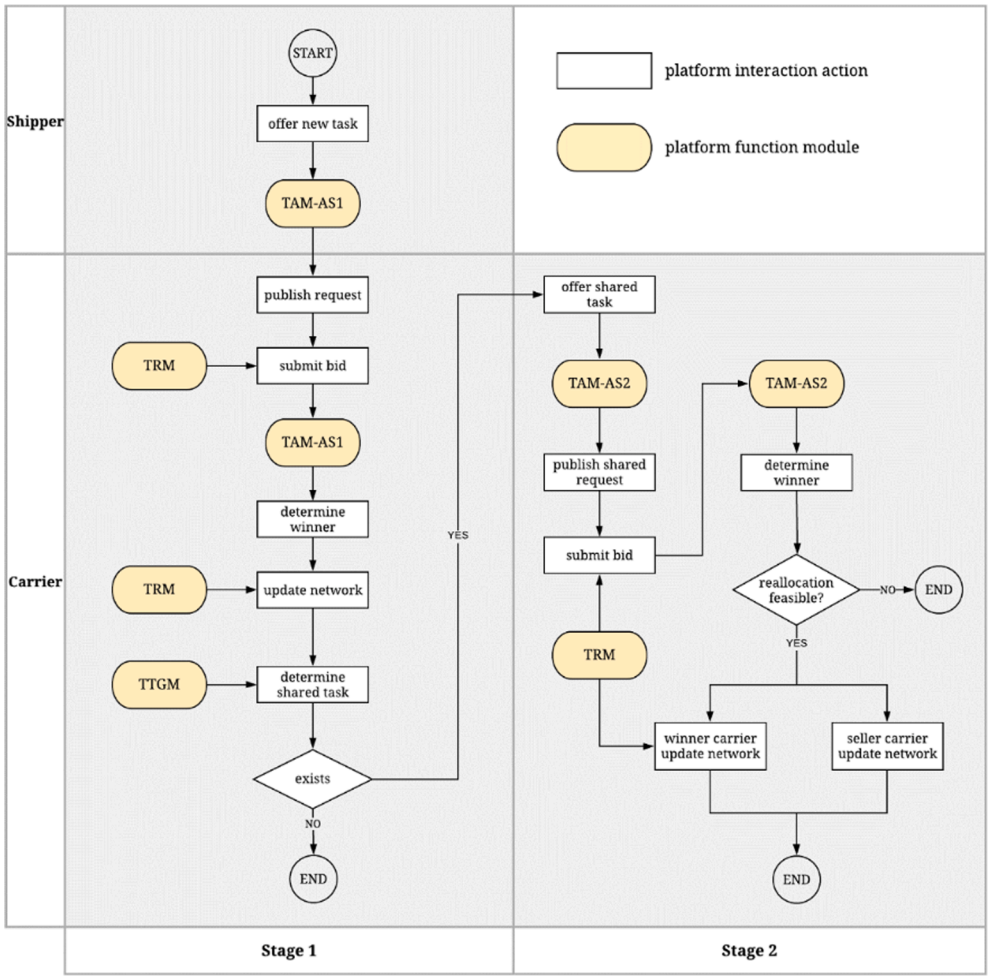


Fig. 3. ACRT framework.

(1) Most research about on-demand logistics has focused on single-dimension collaboration. To the best of our knowledge, there are few studies that have considered combined decision-making for multiple stakeholders (including both shippers and carriers) with multiple objectives for on-demand logistics, using both vertical collaboration and horizontal collaboration.

(2) As described in the literature, auctions could be more efficient and effective in decentralized transportation contexts (Handoko et al., 2014; Stephanus Daniel and Lau, 2016). However, no formal quantitative research into on-demand logistics collaboration has been conducted considering dynamic two-stage auctions, as well as transshipment-based routing simultaneously.

(3) There has been a limited amount of research that has focused on the dynamic pickup and delivery problem with transshipment and time windows (DPDPTTW) in the transportation science and operations research literature. Previous studies have dealt with the static PDPTTW. In addition, a detailed algorithm for the design of transshipment insertion is missing in previous studies.

(4) An integrated and practically usable framework is required. Tactical-level auctioning and operational-level routing decisions should be considered together. The interrelationship between auction-driven decision phases should be explored. For example, how do operational strategies affect the (re)allocation and pricing of auctions, how do auctions affect route optimization, and how does the hierarchy of decision-making become synchronized through the generation and sharing of transshipment-based tasks.

(5) Hybrid B2C & C2C operating models of on-demand logistics have been scarcely analyzed in the literature. In the ACRT presented in this paper, three typical omnichannel scenarios of on-demand logistics are considered: (1) B2C scenario, (2) C2C scenario, and (3) hybrid B2C & C2C scenario. Comparison studies should be designed based on typical operating models under these omnichannel scenarios, through a series of performance criteria, to verify and validate that combined planning and implementation of B2C and C2C hybrid networks is theoretically workable. Moreover, the impacts of key influencing factors on the performance of the ACRT should be investigated to justify the practical significance of proposed solution.

3. Paradigm of ACRT for omnichannel on-demand logistics

A collaborative routing platform can be considered as a service platform for the real-time management of on-demand pickup and delivery tasks, where multiple shippers can trade with multiple carriers. Shippers can be any retailers or individual customers who offer pickup and delivery tasks within the daily time period. Carriers refer to a group of logistics enterprises who can carry out on-demand tasks with sufficient transportation resources. In this service platform, we consider an omnichannel on-demand logistics market within one specific delivery zone in a metropolitan city. Available carriers help shippers deliver goods (e.g., parcels or documents) from origin points to any destination in the prescribed geographical zone.

The TAM enables transportation service trading among platform participants. In the first stage, the TAM primarily minimizes shippers' payments. On-line auction studio 1 (AS1) is implemented for the shippers' transportation procurement. The second stage maximizes carriers' profits and on-line auction studio 2 (AS2) is applied to enable carriers' tasks outsourcing/insourcing. All transactions among participants are realized through either AS1 or AS2. In AS1, each shipper is a seller offering tasks, while each carrier is a bidder. In AS2, each carrier can outsource and insource tasks, their role can be either a seller or a bidder. The platform acts as an auctioneer to determine the winner in each round of the auction. Auction mechanisms are designed for the platform to fulfill these decentralized transportation transactions (Section 4.1).

There are three decision-making problems for each carrier involved in the ACRT. The first one is to determine the uneconomic path to be offered as a shared task for outsourcing. A TTGM is developed, as one of the main platform functions, for each carrier to identify target paths (Section 4.2). The second is route (re)optimization when a carrier acquires new tasks on a real-time basis and the routing problem conforms to the DPDPTTW. The final problem involves determining the bidding price in either AS1 or AS2. Since the bidding price is derived from the operational cost, a routing problem is involved as well. The TRM, as another main platform function, can be used by carriers to make their decentralized decisions on route (re)optimization and to determine their bidding price (Section 4.3).

A typical operation procedure is presented in Fig. 3. Shippers dispatch the on-demand tasks (including location and time information) to AS1. Then AS1 packages the tasks received during an interval and publishes them as a request, and we call it a published request which is consists of a bundle of tasks. Since multiple tasks might be included in a request, the combinational auction (CA) is applied to optimize the allocation of tasks (De Vries and Vohra 2003). We define that any task in the request can be combined into a group, e.g. n tasks of a bundle form $2^n - 1$ groups. Carriers bid for the groups. Available carriers firstly check their operating costs to fulfil each group and submit sealed bids to the platform. The platform determines the winning bidder. When a carrier acquires a group from AS1, their delivery paths will be reconstructed by merging the new task into their dynamic routes. Route (re)planning is based on the real-time information from carriers' mobile devices and digital connectivity. Carriers can dynamically check their uneconomical paths for outsourcing after the delivery route is (re)planned. The identified uneconomical paths will be sent to AS2 as a shared task for outsourcing to other carriers. The shared task consists of location information, lead time and an upper bound price which is associated with the seller carrier's operating cost for fulfilling the shared task by the seller carrier. AS2 packs shared tasks from multiple carriers and publishes them as a shared request, and we call it a published shared request. The auction procedure of AS2 is similar to that of AS1. In the ACRT, an improvement in social welfare of all stakeholders should be maintained.

Regarding network operation, we call both locations of retailer stores and customer sites c-nodes for simplicity. Each carrier is allowed to pick up or drop off parcels of multiple tasks at one c-node simultaneously. Parcels can be transshipped at each c-node. The transshipment-based route planning has two main advantages. First, the transshipment is able to reduce more travel costs compared with non-transshipment planning in the pickup and delivery network of each carrier (Oertel, 2001). Second, transshipments at any pickup/delivery node can enable more flexible task sharing among carriers (Deutsch and Golany, 2018). In the dynamic collaborative network, each task can be formed in either of the two states: before-execution and in-execution. A before-execution task implies the parcel has not been collected. In-execution task means the parcel has been collected from its origin, however, not been delivered, it might be either in a running vehicle or stored at a c-node for transshipment. Both before-execution and in-execution tasks are considered as carriers' existing tasks in their real time network and these are allowed to be outsourced.

Several assumptions are specified to simplify the ACRT:

1) The total transportation resources (freight vehicles) of all carriers are enough to fulfill the total demand in the market.
2) Each carrier has one depot and a limited number of vehicles for delivery. All vehicles belonging to the carriers will start and end at their depots.
3) The capacity of vehicles is not considered since parcels in pickup and delivery logistics usually only require a small fraction of a vehicle's capacity (Berger and Bierwirth 2010). Similarly, the capacity of customer sites for transshipment is ignored.
4) All parcels start/stop at the c-nodes (Hong et al., 2019). Drivers' processing time (e.g., goods (un)loading) at each c-node is ignored.
5) The execution time of auctions for both AS1 and AS2 is ignored.

4. ACRT mechanism design

An integrated mechanism designed for the complicated omnichannel on-demand logistics through transshipment consists of three sub-components is described in this section.

4.1. Two-stage auction mechanism

The following notation is used to formulate the problem:

Indexes	
s	Shipper index
e	Carrier index
Parameters	
Q	Set of groups of a published request/shared request
K_e	Set of existing tasks (including before-execution and in-execution) in the real-time network of carrier e
k_e^s	The shared task whose seller is carrier e
K^u	Set of tasks/shared tasks in a published request/shared request
b_{qe}	Bid price for group q of carrier e
c_{qe}	Marginal cost to serve group q of carrier e
l_{qe}	Marginal tour length to serve group q of carrier e
d_k	Direct tour length of task k from its origin to destination
$L(\cdot)$	Tour length to serve a set of tasks
α_1, α_2	Base rate and distance dependent transportation rate per kilometre
β	Traveling cost per kilometre
p^m	Market price for a task
p^{up}	Upper bound price of a shared task
P_{k^u}	Real payment for the transport service of task k^u
p'_{k^u}	Payment of task k^u based on the bid in auction
p_q^1, p_q^2	First minimum price and second minimum price for group q
Variables	
x_{qe}	1, group q is allocated to carrier e; 0, otherwise.
$y_{k^u e}$	1, task k^u is allocated to carrier e; 0, otherwise.

The objective of AS1 is to minimize shippers' payments for procuring transport services which is formulated in Eq. (1). AS2 aims to maximize carriers' profits and the formulation is given in Eq. (2).

$$Min \sum_{s} \sum_{(k^u \to s)} P_{k^u} \tag{1}$$

$$Max \sum_{e} \sum_{k^u} P_{k^u} * y_{k^u e} - \sum_{e} \sum_{Q} c_{qe} * x_{qe} \tag{2}$$

For AS1, the first price auction mechanism is adopted based on its objective. The carrier who submits the lowest price will be selected as the winner. Delivery services for on-demand tasks are usually within certain areas, thus the service fee fluctuates within a limited range. It has been demonstrated that first-price auction is preferred by all buyers and sellers in low value interval fluctuation auctions (Hu et al., 2010). In addition, first-price auctions are more transparent for both buyers and sellers since the winning bid is the payment price. Truthful bidding can be guaranteed when the fair market price is clear (Maciej Z. 2017). In the ACRT, there could be a single, or more than one task included in a group. We derive the payment price for a certain task from auction p'_{k^u} by splitting the bid price of the selected group according to the proportion of direct tour lengths of each task in its group (see Eqs. (4) and (7)). Usually, the real payment for shipper P_{k^u} is the same as the payment price from auction p'_{k^u}. A restriction is that if p'_{k^u} is higher than the market price, the real payment for the shipper will be equal to the market price. In practice, participant shippers are reluctant to pay a price higher than the current market value. Eqs. (3)–(5) show these constraints.

For AS2, a second price auction mechanism is applied for determining the winner. The second price auction offers incentives to bidders to submit truthful bids and guarantees the request to be allocated to the carrier with the lowest operation cost. With this advantage, it has been adopted in a broad range of auction-based transportation models for carrier collaboration (Andres Figliozzi et al., 2003; Van Duin et al., 2007; Berger and Bierwirth 2010). It should be noted that there is an upper bound price p^{up} for each shared task. The upper bound price is related to the operating cost for fulfilling the shared task by the seller carrier themself. In this research, we let the upper bound price equal the seller carriers' operating cost for the shared task. This indicates the maximum payment for the shared task of the seller carrier is equivalent to their own operating cost. Ideally, all seller carriers expect to pay less for subcontracting. Relevant constraints for the auction of AS2 is described in Eqs. (6)–(10). If p'_{k^u} is lower than the upper bound price, the winning carrier will hand over the shared task, and the real payment for the seller carrier P_{k^u} is the same as p'_{k^u}. If p'_{k^u} is higher than the upper bound price while constraints (9) holds, the real payment is equal to the upper bound price. Otherwise, reallocation of the shared request fails. The two-stage pricing mechanism proposed by Berger and Bierwirth, (2010) is adopted to determine the bid price for the group of tasks which are given in Eqs. (11)–(13). The tour length returned by function $L(\cdot)$ involves the transshipment-based routing mechanism (Section 4.3). Winner determination in the CA is based on Eqs. (14)–(17), where Eq. (14) aims to select the cheapest combination of groups, Eqs. (15)–(16) ensure that no more than one group can be selected from each carrier's bid proposals and each task can be only

allocated to one carrier.

$P_{k^u} = min\{p^m, p'_{k^u}\}$	$\forall k^u \in K^u$	(3)	AS1
$p'_{k^u} = p_q^1 * \frac{d_{k^u}}{\sum_{k' \in q} d_{k'}}$	$\forall k^u \in K^u, q \in Q$	(4)	
$p_q^1 = min\{b_{qe}, \forall e\}$	$\forall q \in Q$	(5)	
$P_{k^u} = \{p^{up}, p'_{k^u}\}$	$\forall k^u \in K^u$	(6)	AS2
$p'_{k^u} = p_q^2 * \frac{d_{k^u}}{\sum_{k' \in q} d_{k'}}$	$\forall k^u \in K^u, q \in Q$	(7)	
$p_q^2 = min\left\{b_{qe}, b_{qe} \neq p_q^1, \forall e\right\}$	$\forall q \in Q$	(8)	
$p_q^1 * \frac{d_{k^u}}{\sum_{k' \in q} d_{k'}} \leq p^{up}$	$\forall k^u \in K^u, q \in Q$	(9)	
$p^{up} = (L(K_e) - L($	$\forall e$	(10)	
$b_{qe} = \alpha_1 + \alpha_2 \times c_{qe}$	$\forall q \in Q, e$	(11)	AS1&AS2
$c_{qe} = \beta \times l_{qe}$	$\forall q \in Q, e$	(12)	
$l_{qe} = L(K_e \cup q) - L(K_e)$	$\forall q \in Q, e$	(13)	
$Min \sum_e \sum_Q b_{qe} * x_{qe}$		(14)	
$\sum_Q x_{qe} \leq 1$	$\forall e$	(15)	
$\sum_e y_{k^u e} = 1$	$\forall k^u \in K^u$	(16)	
$y_{(k^u \in q)e} = x_{qe}$	$\forall q \in Q, e$	(17)	

4.2. Transshipment-based task generation mechanism

Notation is used to formulate the problem:

Index	
o	c-nodes (includes both retailer stores and customer sites)
Parameters	
R	Route set of each carrier on real-time basis
G_r	Candidate path set of routes r
O_{g_r}	Set of c-nodes of path g_r
$f_p^k(o)$	1, parcel of paired task k is picked up from c-node o; 0, otherwise
$f_d^k(o)$	1, parcel of paired task k is delivery to c-node o; 0, otherwise
Variables	
$x(r)$	1, route r is selected; 0, otherwise
$y(g_r)$	1, path g_r of route r is selected; 0, otherwise

As described in Section 3, vehicles visit c-nodes (e.g., stores or customer sites) to execute tasks. Three actions are possible to be undertaken when a vehicle visits a c-node, namely pickup only, delivery only and pickup together with delivery. In addition, parcels to be picked up and delivered can belong to single task or multiple tasks. The proposed mechanism aims to find the most uneconomic path of a carrier's network. 'Uneconomic' is defined as high operative cost which means cutting off the path can decrease the average cost of the route. We define a path as any single or successive c-nodes of the route. Therefore, if a route consists of M c-nodes, the total number of candidate paths of this route is $M + \frac{M(M-1)}{2}$. An example for a simple dynamic route $vehicle - o_1 - o_2 - o_3 - depot$, the set of paths is $\{(o_1), (o_2), (o_3), (o_1, o_2), (o_2, o_3), (o_1, o_2, o_3)\}$. Though we have assumed that the execution time of auctions are ignored in our model for simplicity, it should be noted that the running vehicles keep moving in real-world operations. To avoid some selected c-nodes from being outsourced and meanwhile executed by the carrier themselves, we exclude c-nodes which are approaching the running vehicles when selecting uneconomic paths. For example, in the mentioned route $vehicle - o_1 - o_2 - o_3 - depot$, if o_1 is excluded for checking, the candidate set of paths for outsourcing will be $\{(o_2), (o_3), (o_2, o_3)\}$. As described in Eq. (18), the objective is to maximize the average travel cost saving by excluding the most uneconomic path from the carrier's network.

$$Max: \sum_R \sum_{G_r} \frac{\beta L(\cdot)_{(r)}}{N_{(r)}} x(r) y(g_r) - \sum_R \sum_{G_r} \frac{\beta L(\cdot)_{(r/g_r)}}{N_{(r/g_r)}} x(r) y(g_r) \tag{18}$$

$$\sum_R x(r) \leq 1 \tag{19}$$

$$\sum_{G_r} y(g_r) = x(r) \quad \forall r \in R \tag{20}$$

$$\frac{\beta L(\cdot)_{(r/g_r)}}{N_{(r/g_r)}} x(r) y(g_r) \leq \frac{\beta L(\cdot)_{(r)}}{N_{(r)}} x(r) y(g_r) \quad \forall r \in R, g_r \in G_r \tag{21}$$

Note that (r/g) is route r after deleting path g. Eqs. (19) and (20) are to ensure no more than one path is selected. Eq. (21) describes

1: **FOR** each r in R:
2: Calculate average cost u of r. Where $u = \frac{\beta L(\cdot)_{(r)}}{N_{(r)}}$
3: Generate the candidate path set G_r
4: **FOR** each g_r in G_r:
5: **IF** g_r satisfy function (8) or (9) or (10):
6: Calculate the average cost u' of after excluding path g_r, Where $u' = \frac{\beta L(\cdot)_{(r/g_r)}}{N_{(r/g_r)}}$
7: **IF** $u' < u$:
8: Mark g_r as a possible shared path
9: **End IF**
10: **End IF**
11: **End FOR**
12: **End FOR**
13: Select the path with the most average cost saving from all possible shared paths.
14: **End**

Fig. 4. Greedy search of TTGM.

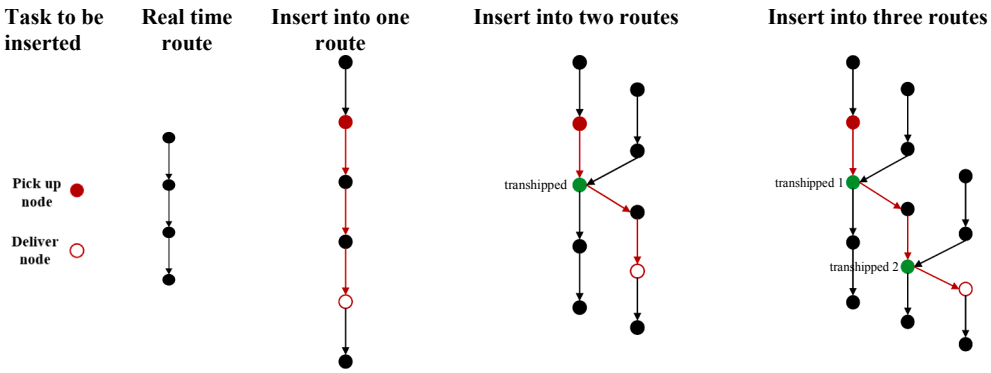


Fig. 5. Transshipment-based task insertion options.

that average cost should be decreased after excluding the path. The feasible paths to be outsourced are classified into three types: pickup paths, delivery paths and independent paths. Pickup paths are to outsource pick up c-nodes. The buyer carrier picks up outsourced parcels and stores them at a c-node as requested by the seller carrier to transship and complete the following delivery, we call this c-node a transshipment c-node. The delivery path is to outsource delivery c-nodes. The seller carrier stores the outsourced parcel at a transshipment c-node along the way, the buyer carrier collects the outsourced parcel from there and finishes the delivery. A paired pickup and delivery task can consist of a pickup or delivery path. An independent path consists of a set of paired pickup and delivery tasks. There is no transshipment c-node connecting the seller carrier and buyer carrier. Eqs. (22)–(24) determine whether a path is feasible to be outsourced, where Eq. (22) formulates the pickup path, Eq. (23) describes the delivery path and Eq. (24) is the independent path. If a path satisfies any of the Eqs. (20)–(22), this path is feasible to be outsourced.

$$f_p^k(o) - f_d^k(o') \geq 0, \sum_O \sum_K f_p^k(o) > \sum_O \sum_K f_d^k(o) \forall o, o' \in O_{g_r}, k \in K \tag{22}$$

$$f_p^k(o) - f_d^k(o') \leq 0, \sum_O \sum_K f_p^k(o) < \sum_O \sum_K f_d^k(o) \forall o, o' \in O_{g_r}, k \in K \tag{23}$$

$$f_p^k(o) - f_d^k(o') = 0 \forall o, o' \in O_{g_r}, k \in K \tag{24}$$

It is obvious that a great number of parameters and variables are involved in the TTGM such as route length of any enumerated route or relationship between each parcel and each node, which are all contributing to the computational complexity if a large-sized problem is considered. Therefore, we use the following greedy search to determine the most uneconomical path (see Fig. 4).

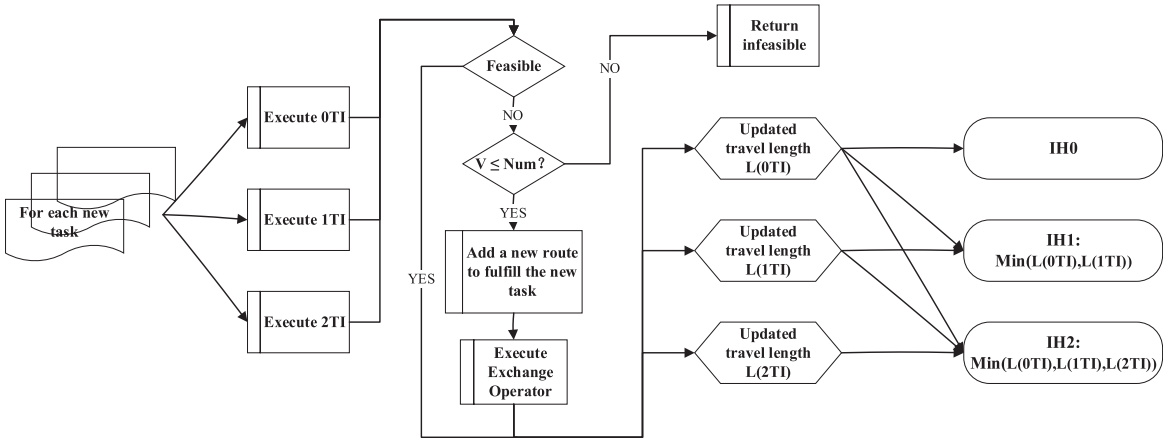


Fig. 6. Framework of the proposed Insertion Heuristics.

4.3. Transshipment-based routing mechanism

Notation is used to formulate the problem in this section:

Parameters	
Num	Number of vehicles available for the carrier
V	Number of vehicles running at real time of the carrier
r'	a new added route
T_{k_p}, T_{k_d}	Real pickup time and delivery time of task $k, k \in K^u \cup K$
w_{k_p}, w_{k_d}	Acquired earliest pickup time and latest delivery time of task $k, k \in K^u \cup K$
Variables	
$z(r)$	1, route r is involved for inserting the task; 0, otherwise
$z(r')$	1, a new route is added; 0, otherwise

Recall that one problem faced by each carrier is to reconstruct their network by arranging the tasks acquired from either AS1 or AS2 to the current routes on a real-time basis. In non-transshipment network planning, it is easy to insert a task into a single route at a proper position. We consider real-time transshipment-based network planning, aimed at inserting each task k^u into the network with a minimum travel cost. There can be more than one route involved in inserting the task as Fig. 5 shows. The optimal task insertion can be obtained via the following program:

$$Min \sum_{R} \beta L(\cdot)_{(r \leftarrow k^u)} z(r) + \beta L(\cdot)_{(r' \leftarrow k^u)} z(r') \tag{25}$$

$$Min|V| \tag{26}$$

$$w_{k_p} \le T_{k_p}, T_{k_d} \le w_{k_d} \quad \forall k \in K^u \cup K \tag{27}$$

$$V \le Num \tag{28}$$

Note that, if inserting task k^u into set R is infeasible, a new route r' can be added to fulfill the task. $(r \leftarrow k^u)$ describes that route r is involved for inserting task k^u. Eq. (25) defines the objective of travel cost minimization by inserting the task. Minimizing the running of vehicles is defined in Eq. (26). Constraint (27) ensures the pickup and delivery should meet the time window of each task. Constraint (28) guarantees the capacity of available vehicles of the carrier is not violated. It is possible that the program returns an infeasible solution based on the restriction of time-windows and limitation of the number of vehicles. Note that, each carrier implements the TRM to check their operational costs for the tasks before auctions. If they receive an infeasible result from the TRM, they will not bid for it and this task will not be assigned to them by auction anyway. It implies all tasks gained from auctions for each carrier must be feasible to be inserted into their network.

Clearly, all pickup and delivery problems are NP-hard (Bouros et al., 2011). It will be extremely complex when considering multiple transshipments. Therefore, an effective heuristic for the DPDPTTW needs to be constructed. A task *Insertion Heuristics* has been developed for the real-time network planning. We examine insertion methods including non-transshipment insertion (0TI), one-transshipment insertion (1TI) and two-transshipment insertion (2TI) when inserting a task. The proposed heuristics include, Insertion with Non-Transshipment (IH0), Insertion with One-Transshipment (IH1) and Insertion with Two-Transshipments (IH2) that are developed based on insertion methods. Fig. 6 outlines the framework of the related heuristics and insertion methods. It can be concluded that the general concept of *Insertion Heuristics* with N transshipments (IHN) is that for the task, trying multiple options to insert it, where the possible insertion methods starts from inserting it without transshipment (achieved by 0TI) to inserting it with N

```
FOR each r in R:
    Insert task k^u into r with minimize travel length and no violate of each task's time window
End FOR
IF there are feasible inserted solutions:
    Select the best solution
ELSE:
    Add a new route
    Execute Exchange Operator
End IF
```

0TI

```
FOR each r in R:
    Insert the pickup node of task k^u into r with minimize travel length and no violate of each task's time window. Derive the inserted position ip
    For each customer node o from position ip + 1 to the end of r:
        Let node o be the transfer node, and generate a new pair task k_new with o as origin and task k's delivery node as destination
        Execute 1TI to insert k_new to R\r
    End FOR
    IF there are feasible inserted solutions:
        Select the best solution
    End IF
End FOR
IF there are feasible inserted solutions:
    Select the best solution
ELSE:
    Return infeasible
End IF
```

2TI

```
FOR each r in R:
    Insert the pickup node of task k^u into r with minimize travel length and no violation of each task's time window. Derive the inserted position ip
    For each customer node o from position ip + 1 to the end of r:
        Let node o be the transfer node, and generate a new pair task k_new with o as origin and task k's delivery node as destination
        Execute 0TI to insert k_new to R\r
    End FOR
    If there are feasible inserted solutions:
        Select the best solution
    End IF
End FOR
IF there are feasible inserted solutions:
    Select the best solution
ELSE:
    Return infeasible
End IF
```

1TI

```
FOR each r in R:
    FOR each before executed task k^r of route r:
        Insert k^r into the new added route in the proper position
        IF insertion is feasible and result in cost saving:
            Keep the exchange
        Else:
            Undo the exchange
        End IF
    End FOR
End FOR
```

Exchange Operator

Fig. 7. Solution procedures for the examined insertion methods and exchange operator.

transshipments (achieved by NTI), selecting the best option as the solution. Computational analysis for the proposed heuristics are presented in Appendix B. It was found that with the increased number of transshipment attempts, the computational time (CPU time) increases substantially due to the complexity of the calculations. As real time management for on-demand logistics is time sensitive, therefore, we investigated no more than two transshipments when inserting a task in this section. The implementation procedures for 0TI, 1TI, 2TI and the Exchange Operator outlined in Fig. 7.

5. Computational study

This section aims to verify the performance of the ACRT. Additionally, sensitivity analyses are conducted to investigate the impacts of various key factors.

5.1. Experimental setup

Computational experiments were conducted on a HP EliteBook with an Intel core i7 2.70 GHz processor and 16 GB RAM. Programming of the simulation of the TAM, TTGM and TRM was completed in Python 3.7. The CA that determines task (re)allocation for

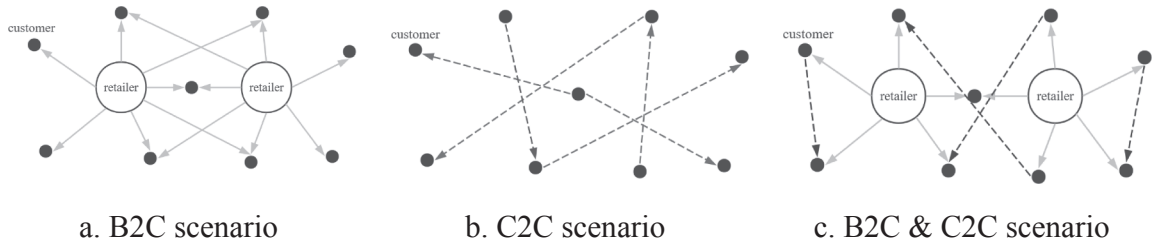


Fig. 8. Network structure for omnichannel on-demand logistics scenarios.

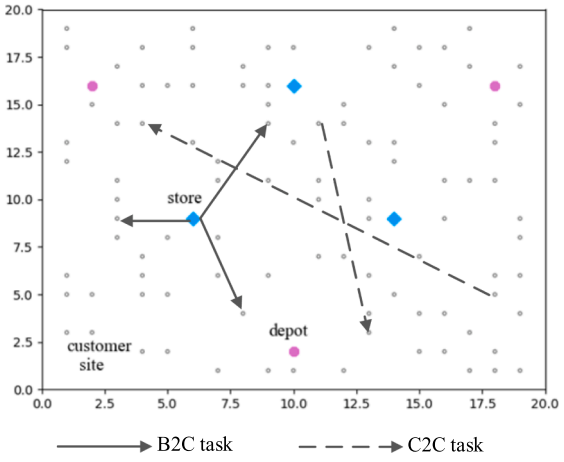


Fig. 9. Scheme of the study area.

the TAM was solved by the CPLEX solver. In order to speed up the simulation for instances with a large number of tasks, we used the speed of vehicle v as a simulation speed where 1 h of real-time operation is simulated in $1/v$ hours of computer time (Mitrović-Minić and Laporte 2004).

In the experiments, our primary focus was to verify the performance of the ARCT by comparing it with different models. The models examined include:

a) **Random model**, which is close to the conventional mode. The tasks arriving from shippers are distributed to carriers randomly, while carriers carry out a separate network.
b) **AS1 model**, in which AS1 is implemented to optimize freight transportation procurement between shippers and carriers.
c) **Auction-enabled collaborative routing model (ARC)**, in which both AS1 and AS2 are implemented, however, the routing mechanism and the shared task generation mechanism for AS2 is based on a non-transshipment network.
d) **ARCT model,** in which both AS1 and AS2 are implemented. The routing mechanism and shared task generation mechanism for AS2 is based on a transshipment network.

Three omnichannel scenarios are considered for the comparison: (1) B2C scenario, (2) C2C scenario, (3) hybrid B2C & C2C scenario. For the B2C scenario, on-demand delivery tasks are offered by retailers (shippers). Carriers pick up parcels from retailers' stores and deliver to consumer sites based on the prescribed order information. Considering the online order return policy, a small fraction of return deliveries (pick up from the receiver and deliver back to the retailer) are also considered in this scenario. The structure of B2C scenarios are a satellite network where different tasks are initiated from the same retailer and completed at different sites of end consumers, as shown in Fig. 8a. The profile of the C2C scenario is a horizontal network where tasks are offered by individual customers, the origin and destination of tasks are randomly distributed over the studied area (Fig. 8b). A hybrid network incorporating B2C and C2C is illustrated in Fig. 8c.

We consider an area covering 20 km × 20 km with three retailers (shippers), three logistic companies (carriers), and one hundred customer sites. Each retailer has an off-line store which is also the pickup location of the on-demand tasks. Each carrier has their own depot, all vehicles start and end at such a depot. The location of the retailers' stores and carriers' depots are evenly distributed over the area, while the location of 100 customer sites are randomly generated. Fig. 9 shows the configuration of the study area and several examples of tasks. We use linear distances for simplicity.

The main performance criterion is the social welfare change between two compared modes, denoted by $M1$ and $M2$:

Table 1
Experimental parameters.

Parameters	Values	Notes
Base rate per task: α_1	3 (direct tour length of task ≤3 km) for AS1 5 (direct tour length of task >3 km) for AS1 0 for AS2	Empirical assumption based on the average market situations
Distance dependent transportation rate per kilometre: α_2	3 for AS1; 1 for AS2	
Operating cost per kilometre in urban area (consist vehicle and employee): β	1	
Market price for conventional mode: p^m	\$8 (direct tour length of task ≤3 km) \$20 (direct tour length of task >3 km)	
Speed of vehicle:v	45 km/h	
Number of vehicles available for each carrier :Num	5	

$$\Delta W_{(M1,M2)} = \delta_s \cdot \Delta SW_{(M1,M2)} + \delta_c \cdot \Delta CW_{(M1,M2)} + \delta_r \cdot \Delta RW_{(M1,M2)} \tag{29}$$

$$\Delta SW_{(M1,M2)} = \Delta\left(\sum_{s}\sum_{(k^u \to s)} P_{k^u}\right)_{(M1,M2)} \tag{30}$$

$$\Delta CW_{(M1,M2)} = \Delta\left(\sum_{e}\sum_{(k^u \to e)} P_{k^u} - \sum_{e}\sum_{(q \to e)} c_q\right)_{(M1,M2)} \tag{31}$$

$$\Delta RW_{(M1,M2)} = \Delta(0.025 \times 62L)_{(M1,M2)} \tag{32}$$

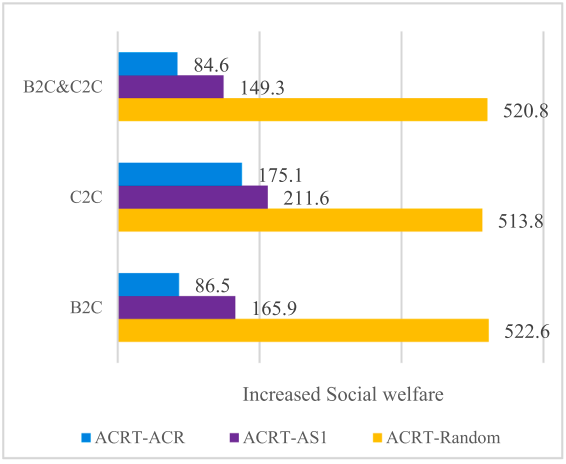
where $\Delta SW_{(M1,M2)}$ is shippers' welfare change, $\Delta CW_{(M1,M2)}$ is carriers' welfare change, and $\Delta RW_{(M1,M2)}$ is residents' welfare change. The parameters $\delta_s, \delta_c and \delta_r$ are the welfare weights which are set equal to one (Rouhani et al., 2016). More specifically, $\Delta SW_{(M1,M2)}$ is the gap of shippers' total payments between compared modes (defined in Eq. (30)), $\Delta CW_{(M1,M2)}$ is the gap of carriers' total profit (defined in Eq. (31)), and $\Delta RW_{(M1,M2)}$ is the gap of CO_2 emission costs (defined in Eq. (32)). Where unit emissions are \$25/ton of CO_2 (Rouhani et al., 2016) and CO_2 emissions are defined as $E = 62L$, where L is the tour length in kilometers (Cefic., 2011).

The reported solutions are averages of 20 randomly generated instances. For the B2C scenario, the origin of a task is randomly generated from the three retailers, its destination is randomly generated from the 100 customer sites. We set the return rate of online orders to be 0.2 based on an empirical assumption (Pei and Paswan ,2018). Thus, 120 tasks are comprised in each instance, consisting of 100 forward delivery (from retailer to customer) and 20 reverse delivery (from customer to retailer) tasks. For the C2C scenario, both the origin and destination of a task are randomly generated from the 100 customer sites. Note that the location of a task's origin and destination cannot be overlapped. Similarly, there are 120 tasks for each instance. For the B2C & C2C hybrid scenario, a total of 120 tasks are randomly generated, including 60 delivery tasks (C2C scenario), 50 forward deliveries (B2C scenario), and 10 reverse deliveries (B2C scenario). Task arrivals are assumed to follow a Poisson process, $P(k) = \frac{\lambda^k e^{-\lambda}}{k!}$, where λ is the average number of tasks revealed in each auction interval. We assume the auction interval is 0.05 h and set $\lambda = 1$. The set of parameters involved in running the experiment are outlined in Table 1.

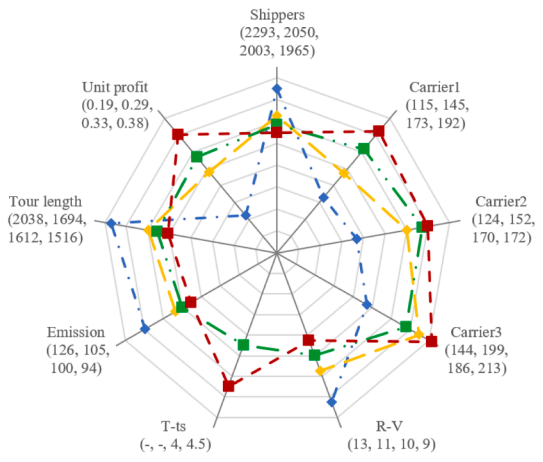
5.2. Comparison among different models

We summarize the findings related to the performance of the examined four models under three omnichannel scenarios in Fig. 10. The results presented are the average results of 20 instances with each consisting of 120 tasks. Besides the main criteria of social welfare gaps, the profit of each carrier $P^{carrier1}, P^{carrier2}, P^{carrier3}$, total tour length L, profit per tour length, actual utilization of vehicles and transshipment times are outlined to give a comprehensive comparison. Fig. 10a. shows that social welfare can achieve a positive improvement in the ACRT compared with the other three models. The most significant improvement exists in contrast to the random model, achieving approximately \$510. The radar charts of Fig. 10b–d. illustrate that the ACRT performs superior to other models in almost all criteria of the three tested scenarios. It can be found that more successful outsourcing happened in the ACRT than that in the ACR especially for the C2C scenario (7.7 times vs 0.8 times) due to flexible freight transshipment. In addition, profit fluctuates greatly over different carriers in the AS1 model, e.g. in C2C scenario, carrier 1′s profit is up to \$330 while carrier 3′s profit is only 267. However, profits can be evenly distributed among multiple carriers in the other models. It indicates that by implementing AS1 only, it is possible for it to lead to unbalanced competition in marketplace.

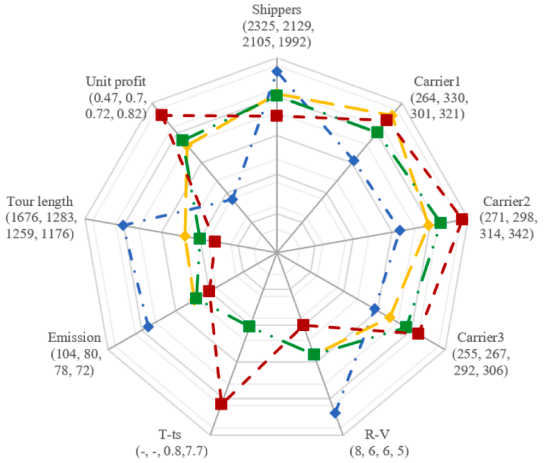
More specific results of each instance are presented in Fig. 11. Shippers' payments within the ACRT are the lowest (as illustrated in Fig. 11a.1–a.3) while carriers' overall profits in the ACRT are the highest (as shown in Fig. 11b.1–b.3) in the four models. The reason for the increased welfare among stakeholders is that the total tour length of the ACRT is the shortest as Fig. 11c.1–c.3 shows. Therefore,



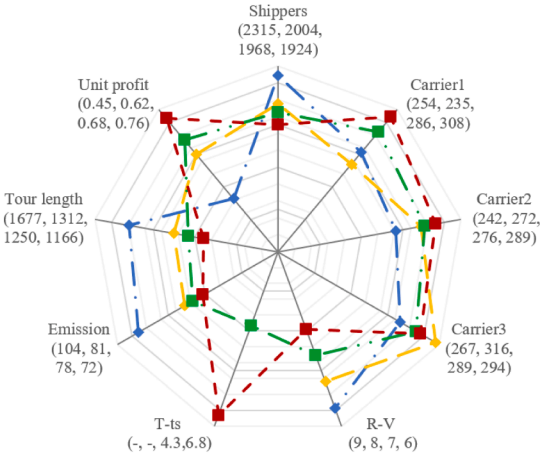
a. Improvement of social welfare



b. Comparison of B2C scenario



c. Comparison of C2C scenario



d. Comparison of B2C&C2C scenario

Legend:

Random AS1 ACR ACRT

Carrier: profit ($) *Shipper: payment ($)* *Unit profit: profit per length ($/km)*
Tour length: (km) *Emission: (kg)* *R-V: real number of vehicles used*
T-ts: times of outsourcing between carriers

Fig. 10. Average performance of 20 instances for the models examined.

the unit profit (profit per tour length) of the ACRT is higher than that of other models which is illustrated in Fig. 11d.1–d.3 owing to the reduced tour length, CO_2 emissions and the number of actual vehicles used are reduced in the ACRT. A comparison can be found in radar charts of Fig. 10. The above analysis implies that the ACRT is promising to improve overall social welfare for the omnichannel on-demand logistics market.

Fig. 12 illustrates the workflow of a real-time on-demand task (re)allocation under the four models, captured from the tested instances. In the random model (Fig. 12a), the new tasks arriving from the shipper is allocated to a carrier randomly (carrier 1 is selected in this instance). Since the direct tour length of the task is 12 km which is longer than 3 km, the shipper should pay $20 for the transport service based on our parameter settings. The operation cost of carrier 1 for fulfilling this task is $17.74 and total tour length increases by 17.74 km. In the AS1 model (Fig. 12b), three carriers bid for the task through the on-line auction with bidding prices of $58.23, $8.84 and $35.86, respectively, based on their costs. Carrier 2 finally wins the task and is paid $8.84 according to the First Price auction setting. The ACR enables task outsourcing among carriers. Since this model considers a non-transshipment network, parcels must be processed by the same freight vehicle. Therefore, the shared task should be paired with its origin and destination. As illustrated in Fig. 12c, the identified uneconomic task to be outsourced is the fifth task of carrier 2 acquired from AS1 which is marked with its origin (5O) and destination (5D) as red nodes. An upper bound price of $12.12 is attached with the shared task since the operation cost to fulfil it by carrier 2 is $12.12. The operation cost for processing the shared task of both carrier 1 and carrier 3 are $12.52 and $20.17,

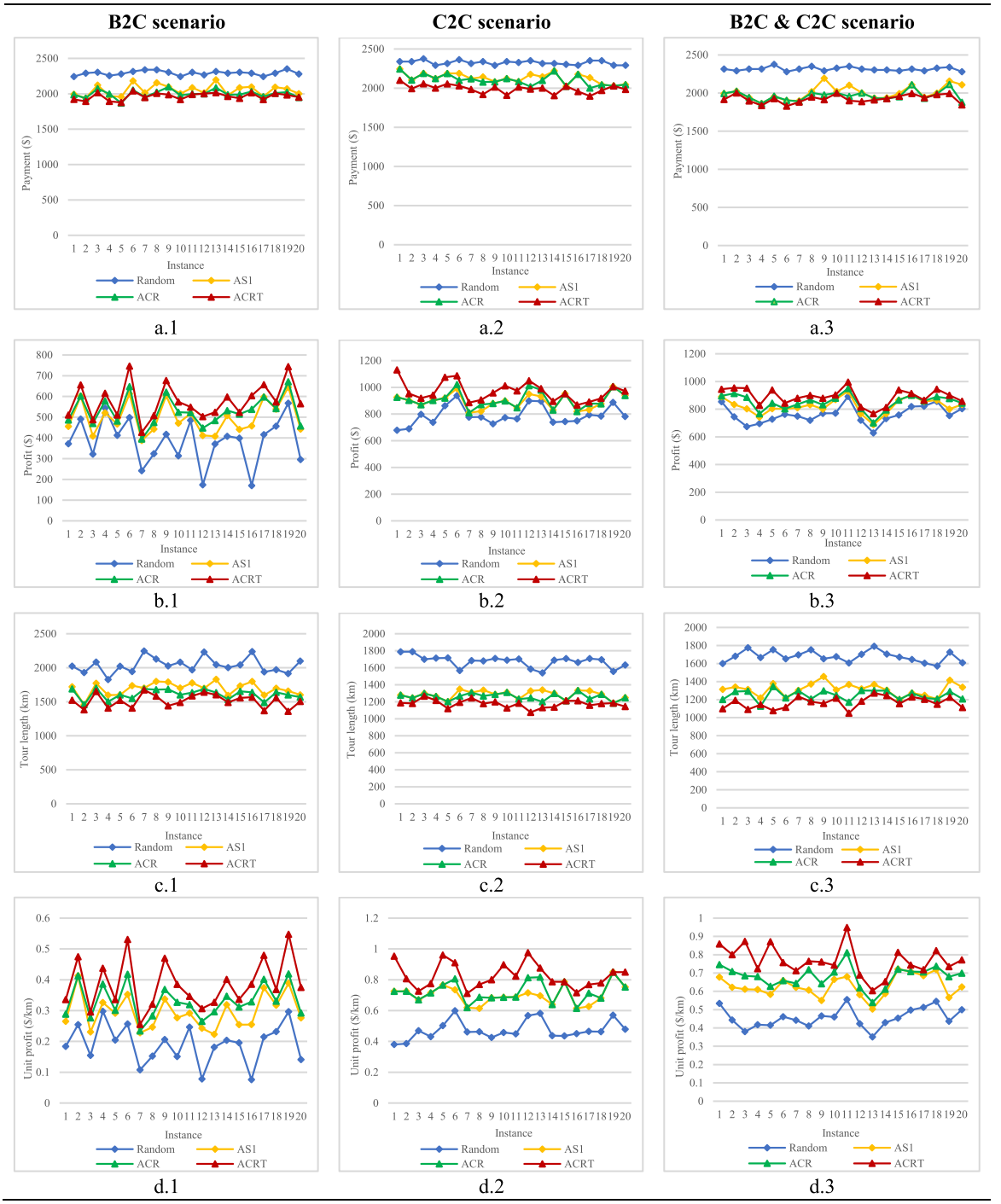


Fig. 11. Detailed performance of the 20 instances for the models examined.

respectively. Therefore, their bidding prices must be higher than $12.12. As a consequence, the auction in AS2 fails and the shared task is still processed by carrier 2. In the ACRT (Fig. 12d), the identified shared task is a path of carrier 2′s network which is marked by the red line in Fig. 12d. The parcel of the fifth task is picked up by carrier 2 and transshipped at a c-node (marked as T), the following delivery path is outsourced. Carrier 1 wins this shared task with a lowest bidding price of $6.31. Since the second lowest bidding price is higher than the upper bound price, carrier 1 should be paid the upper bound price by carrier 2. The successful outsourcing in AS2 reduces the total tour length by 2.42 km.

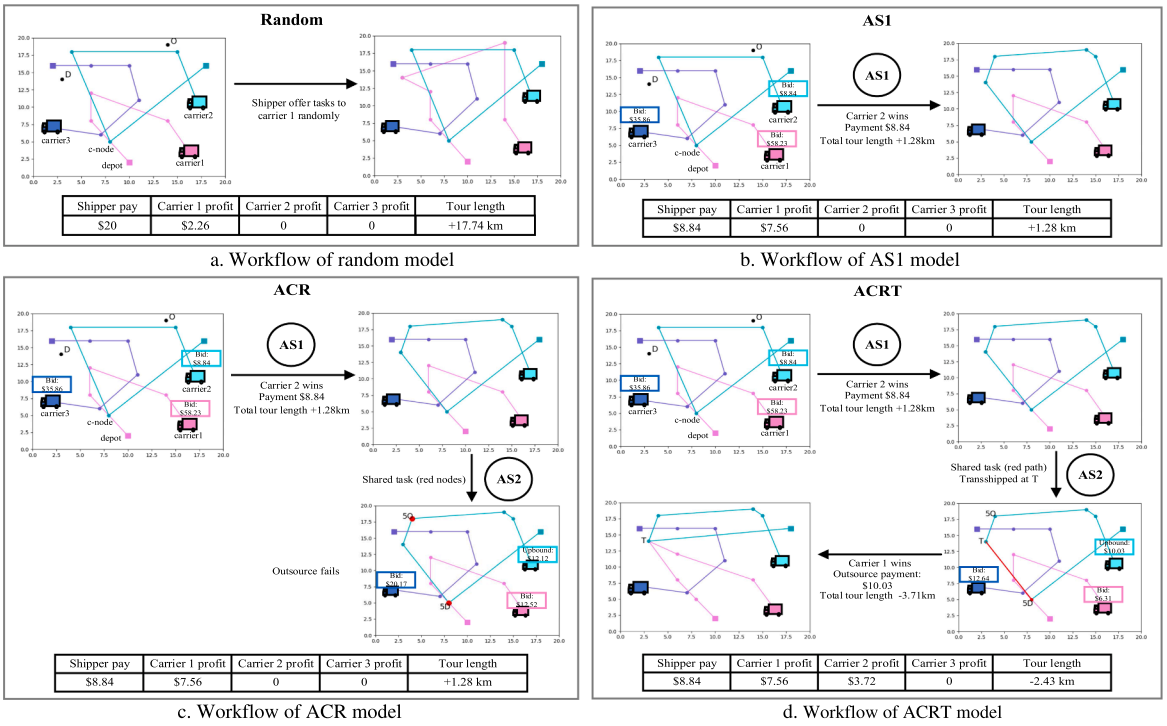


Fig. 12. Workflow of real-time on-demand task (re)allocation under four models.

5.3. *Sensitivity analysis*

In this section, the performance of the ACRT is investigated under the hybrid B2C & C2C scenario, with the following three dimensions: (i) demand density growth during auction interval; (ii) urgency of tasks; (iii) flexible auction intervals. Since task arrivals conform to the Poisson process, $P(k) = \frac{\lambda^k e^{-\lambda}}{k!}$ based on our assumption, where, λ is the average number of tasks arriving in each auction interval. Increasing the value of parameter λ implies a higher frequency of tasks arrival during each auction interval and considering another perspective of growing demand density in market. Performance of the ACRT is examined under different demand density levels. For each 120-tasks instance, arrivals are assumed to be according to the $P(1)$, $P(2)$, $P(3)$ and $P(5)$ distributions, respectively. Section 5.3.1 summarizes the related impacts.

On-demand deliveries are highly time-sensitive where the service must be fulfilled in hours rather than days and the due time of delivery is critical. In Section 5.3.2, instances with different due time tasks $\{1h, 3h, 5h\}$ are examined.

In the above experiments, the ACRT is set with a fixed auction interval of 0.05 h. Section 5.3.3 studies the performance of the ACRT with flexible auction intervals of $\{0.05h, 0.1h, 0.15h\}$. In addition, more tasks have been generated randomly for this investigation. Task arrivals still conform to the $P(1)$ process.

5.3.1. *Impact of demand density growth during auction interval*

Table 2 shows that improvements in social welfare is varied under different demand density settings where improvement hits the peak with the $P(3)$ arrival process. The saving payment of shippers for the 120 tasks increases steadily with the growth of demand density. For example, the total payment in the ACRT is \$1945.5 under $P(1)$ arrival process and decreased to \$1,443.5 under the $P(5)$ arrival process. This is because a significant reduction of tour length (from 1198.2 km to 684.9 km) where vehicle routing can be planned in a more optimal means based on the consolidation synergy. In contrast, the total service payment has not changed with the different demand density in the random model, though the total tour length has been decreased owing to the consolidation synergy (from 1668.4 km to 1085.8 km), however, it scarcely contributes to welfare improvement for shippers. The reason for this is the transportation service fee is fixed in the market regardless the fluctuation of operating costs. The total profit of carriers rises stably with the growth in demand density in the random operative mode because of the unchanged income (payment) as well as the decline of cost (tour length). However, in the ACRT, the total profit increased by \$52.4 from $P(1)$ to $P(2)$arrival process. Then it drops slightly when parameter λ is larger. The reason is that profit is associated with cost (tour length), while income (price) is based on tour length (see Eqs. (10)–(12)); therefore, the decrease of total payment and tour length results in a reduction in profit. However, it can be found that the unit profit (profit achieved per tour length) increases steadily with the growth of demand density which ranges from 0.8 to 1.3 \$/km under tasks arrival processes from $P(1)$ to $P(5)$. In addition, the unit profit from the ACRT is higher than that of the random model for all demand densities. Unsurprisingly, more vehicles will be used to fulfill the 120 tasks when the demand density grows under these two models. More effective sharing/outsourcing are undertaken among carriers with the growth of demand density. This is because

Table 2
Performance of the ACRT with growth of demand density.

		Social welfare measure ($)	Shippers' payment ($)	Carriers' profit ($)	Emission (g)	Tour length (km)	Unit profit (profit per length $/km)	R-V*	T-ts*
P (1)	**Ran***	+489.2	2318.4	790.7	103440.8	1668.4	0.5	8.3	
	ACRT		1945.5	906.3	74290.3	1198.2	0.8	5.7	8.3
	Gap		↓ *16.08%*	↑ *115.6*	↓ *28.18%*	↓ *470.2*	↑ *0.3*	↓ *30.95%*	
P (2)	**Ran***	+637.1	2314.2	905.4	82813.4	1335.7	0.7	9.5	
	ACRT		1731.1	958.7	56221.8	906.8	1.1	6.8	18.6
	Gap		↓ *25.20%*	↑ *53.3*	↓ *32.11%*	↓ *428.9*	↑ *0.4*	↓ *28.34%*	
P (3)	**Ran***	+710.8	2320.5	988.3	74691.4	1204.7	0.8	10.4	
	ACRT		1569.6	947.6	49182.3	793.3	1.2	7.2	19.8
	Gap		↓ *32.36%*	↓ *40.7*	↓ *34.15%*	↓ *411.4*	↑ *0.4*	↓ *30.84%*	
P (5)	**Ran***	+605.6	2311.2	1162.4	67319.6	1085.8	1.1	10.8	
	ACRT		1443.5	899.7	42462.4	684.9	1.3	7.3	22.1
	Gap		↓*37.54%*	↓*262.7*	↓*36.92%*	↓*400.9*	↑*0.2*	↓*31.98%*	

* Ran: Random model; T-ts: times of outsourcing between carriers; R-V: real number of vehicles used.

Table 3
Performance of the two modes under varied due-time settings.

Due time/Urgency level		Social welfare measure ($)	Shippers' payment ($)	Carriers' profit ($)	Emission (g)	Tour length (km)	R-V*	T-ts*
1h	**Ran***	+499.1	2307.5	89.8	144999.4	2338.7	12.09	
	ACRT		2142.2	422.8	112846.2	1820.1	8.27	4.9
	Gap		*↓165.3, 7.2%)*	*↑333.0*	*↓32153.2*	*↓518.6*	*↓(3.82, 31.6%)*	
3h	**Ran***	+510.6	2311.2	749.1	103974.0	1677	8.09	
	ACRT		1963.4	911.1	73662.2	1188.1	5.73	7.3
	Gap		*↓(347.8, 15.0%)*	*↑162.0*	*↓30311.8*	*↓488.9*	*↓(2.36, 29.2%)*	
5h	**Ran***	+458.3	2306.4	1030.4	87376.6	1409.3	7.09	
	ACRT		1853.2	1034.8	60512.0	976.0	4.64	10.9
	Gap		*↓(453.2, 19.6%)*	*↑4.4*	*↓26864.6*	*↓433.3*	*↓(2.45, 34.6%)*	

* Ran: Random model; T-ts: times of outsourcing between carriers; R-V: real number of vehicles used.

when a carrier has more tasks to do, the network can be more redundant. Thus, there will be more opportunities to find uneconomical paths.

5.3.2. Impact of on-demand task urgency

Table 3 shows that social welfare can be improved to fulfill various types of urgent tasks when implementing the ACRT. In addition, with longer tasks' due time, shippers can save more costs in the ACRT compared with the random model. The savings gap of payment starts from 7.2% with due time of 1 h and increases to 19.6% with due time of 5 h. When the due time of tasks is shortened (e.g., tasks must be delivered within 1 h), the profit of each carrier in the ACRT is dramatically higher than that for the random model, the improved gap ranges from $100 to $120. This result is quite realistic that when each carrier is dealing with a bunch of tasks with tight due times separately, the total tour length will be much longer owing to the urgency of each task and the limited facilities available. However, delivery trips can be reduced by carriers' collaboration through the outsourcing/insourcing of tasks in the ACRT. The savings gap of tour length increases with the increased urgency of tasks. It is understandable that when there is enough execution time for the tasks, each carrier is able to optimize their network independently for cost savings, however, independent planning still unable to reach the performance of the collaborative management of the ACRT. Also, in terms of vehicle requirements, the ACRT can reduce the number of actual vehicles used by around 30% compared to the random model in all the instances studied. Clearly, the ACRT has superior performance for each criterion under the examined contexts, owing to transshipment-based collaboration among carriers.

5.3.3. Impact of flexible auction interval

In Table 4, we examined the performance of the ACRT via a fixed auction interval and flexible auction interval. The performance of random model has also been compared for each instance. It is clear that the ACRT performs better for all criteria involving social welfare improvement, carriers' payment, shippers' profit, greenhouse emissions and actual utilization of vehicles than the random model. The number of actual vehicles used is almost kept the same when fulfilling more tasks over time. It implies that the ACRT has robust performance in dynamic operating circumstances. It can be noticed that the ACRT with flexible auction interval achieves better performance than that with fixed auction interval in all cases. For example, in the 200-tasks instances, the ACRT with flexible auction interval achieves an improved social welfare of $177.3, reduction in shippers' payment of $127.7, increase in carriers' profits of $49.2, reduction in CO_2 emissions of 8.9% and decrease in tour length of 220.6 km compared with the fixed auction interval. Additionally, slight reductions of the actual vehicle's usage and effective sharing among carriers have been demonstrated. The reason might be that a flexible auction interval can increase the size of published request bundle; therefore, consolidation synergy works. This result also suggests that especially in off-peak hours, auction intervals can be adjusted a bit longer to enable a more consolidation planning for the entire system.

6. Conclusions

In this article, we investigate collaborative mechanisms for omnichannel on-demand logistics in real-time management. We develop an ACRT which considers an online platform to enable collaboration for each stakeholder such as shippers and carriers. A two-stage combinational auction mechanism is designed for dynamic on-demand tasks. A first price auction is adopted between shippers and carriers to minimize shippers' payment. Task reallocation and sharing among carriers is then fulfilled in second price auction which aims to improve carriers' profit. Other operational mechanisms are designed as decision support tool for carriers. The transshipment-based task generation method is used to identify uneconomical paths from a carrier's real-time network. Based on the updating results of network (re)construction, each carrier can offer the selected path for outsourcing. A transshipment-based routing algorithm is developed to support each carrier for real-time vehicle route planning. The designed tactical-level auctioning and operational-level routing decisions are thus considered together. The operational strategy will affect the (re)allocation and pricing of auction through the bid price, and the auction efficiency will pose direct impact on the result of route (re)optimization. Through the

Table 4
Performance of ACRT with fixed and flexible auction interval.

		Social welfare measure ($)		Shippers' payment ($)		Carriers' profit ($)		Emission (g)		Tour length (km)		R-V*	T-ts*
120	**Ran***			2315.5		767.95		103983.3		1677.2		8.18	
	ACRT-fix	+515.8		1924.5		891.95		72319.9		1166.5		5.82	6.8
	ACRT-flexible	+647.3	↑131.5	1814.7	↓*109.8*	913.4	↑*21.45*	61832.6	↓*114.5%*	997.3	↓*169.2*	4.55	10.4
200	**Ran***			3868.7		1448.7		207898.4		3353.2		8.23	
	ACRT-fix	+724.2		3241.5		1544.3		153412.8		2474.4		5.90	8.8
	ACRT-flexible	+901.5	↑177.3	3113.7	↓*127.7*	1593.5	↑*49.2*	139735.6	↓*8.9%*	2253.8	↓*220.6*	4.65	14.3
400	**Ran***			7625.7		3168.5		301481.2		4862.6		8.27	
	ACRT-fix	+1624.9		6210.2		3375.6		205666.4		3317.2		5.90	18.8
	ACRT-flexible	+1906.1	↑281.2	6001.1	↓*209.1*	3447.2	↑*71.6*	187847.6	↓*8.7%*	3029.8	↓*287.4*	4.9	25.4

* Ran: Random model; T-ts: times of outsourcing between carriers; R-V: real number of vehicles used.

generation and sharing of transport tasks dynamically, the hierarchical decision-making is synchronized.

There are three main contributions made by this article: (1) this article is among the first to improve the overall social welfare involving welfare of shippers, carriers and city residents simultaneously in the collaborative transportation context. Driven by technology developments such as mobile devices and digital connectivity in transportation, we address the dynamic management for complicated omnichannel on-demand logistics. (2) This article develops a set of effective mechanisms for the ACRT. The TAM aims to decrease shippers' payment as well as increase carriers' profits through effective task (re)arrangement. The TTGM identifies the most uneconomic path of each carrier's network efficiently and the TRM contributes to the DPDPTTW. (3) We conducted a computational study to demonstrate the positive impact of social welfare achieved by the ACRT compared with several typical operating models, through the simulated omnichannel scenarios like business-to-customer (B2C) fulfillment, customer-to-customer (C2C) fulfillment and hybrid fulfillment of B2C and C2C. Results show that a positive improvement in social welfare can be achieved by the ACRT compared with other models. It also indicates the combined planning and implementation for B2C and C2C hybrid networks is theoretically workable, which is also consistent with the concept of Physical Internet-enabled hyperconnected city logistics. From the sensitivity analyses, we found that the welfare of shippers has been optimized significantly by the ACRT with the growth of demand density. It also suggests that carriers' collaboration can generate more benefits when fulfilling tasks with tight due times. In addition, it would be more appropriate if auction the interval of the ACRT is adjusted flexibly based on real-time market demand.

Some limitations of this work should be noted. As far as the models are concerned, the bidding prices are determined following the two-stage pricing mechanism from Berger and Bierwirth (2010) for each carrier. However, in reality, the bidding strategy for price determination should be more flexible. Therefore, regarding future work, one critical extension of the ACRT is to explore a more flexible data-driven pricing strategy based on dynamic information or even sophisticated carrier behavior. Another promising line of investigation relates to designing a capacity-constrained mechanism, although we relaxed this constraint in the current study. It would be also desirable to investigate how the concept of the ACRT could be applied to other transportation settings such as less-than-truckload intercity logistics.

CRediT authorship contribution statement

Chaojie Guo: Conceptualization, Software, Data curation, Validation, Writing - original draft. **Russell G. Thompson:** Writing - review & editing, Methodology, Supervision. **Greg Foliente:** Visualization, Writing - review & editing. **Xiang T.R. Kong:** Conceptualization, Methodology, Writing - original draft, Writing - review & editing.

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Appendix A

List of Abbreviations	
ACRT	Auction-enabled collaborative routing mechanism with transshipment
TAM	Two-stage auction mechanism
TTGM	Transshipment-based task generation mechanism
TRM	Transshipment-based routing mechanism
CA	Combinational Auction
AS1	On-line auction studio 1
AS2	On-line auction studio 2
ACR	Auction-enabled collaborative routing model (non-transshipment network)
DPDPTTW	Dynamic pickup and delivery problem with transshipment and time windows
PDPTTW	Pickup and delivery problem with transshipment and time windows
DPDP	Dynamic pickup and delivery problem
B2C	Business-to-Customer
C2C	Customer-to-Customer

Appendix B

The evaluated fifty instances are randomly generated based on the work of Mitrović-Minić and Laporte (2004). Each instance consists of 40 tasks. For each instance the first thirty tasks are used as the warm-up period. The performance of the thirty-first tasks were recorded, including increased tour length of inserting the task and CPU calculation time. Table A lists the results of the tested twenty instances and the entire performance of the fifty instances are illustrated in Fig. A.

Table A
Results of 20 instances.

Increased tour length(km)	0TI	1TI	2TI	IH0	CPU time (S)	IH1	CPU time (S)	IH2	CPU time (S)
1	27.59	9.57	89.13	27.59	*0.006*	9.57	0.054	9.57	0.263
2	42.07	34.31	8.77	42.07	*0.001*	34.31	0.036	8.77	0.336
3	3.08	0.18	56.36	3.08	*0.007*	0.18	0.018	0.18	0.226
4	3.22	64.08	1.07	3.22	*0.001*	3.22	0.008	1.07	0.524
5	9.74	5.69	4.37	9.74	*0.002*	5.69	0.045	4.37	0.458
6	48.41	41.89	38.04	48.41	*0.001*	41.89	0.026	38.04	0.365
7	48.71	72.08	20.07	48.71	*0.001*	48.71	0.010	20.07	0.445
8	40.40	35.40	30.23	40.4	*0.003*	35.4	0.058	30.23	0.364
9	68.91	1.12	0.71	68.91	*0.001*	1.12	0.026	0.71	0.420
10	27.79	4.18	0.95	27.79	*0.004*	4.18	0.053	0.95	0.485
11	3.15	54.29	0.97	3.15	*0.002*	3.15	0.008	0.97	0.367
12	73.89	0.99	0.25	73.89	*0.001*	0.99	0.036	0.25	0.478
13	65.07	9.39	9.23	65.07	*0.001*	9.39	0.041	9.23	0.352
14	42.02	22.54	17.85	42.02	*0.003*	22.54	0.024	17.85	0.326
15	14.16	0.02	0.06	14.16	*0.005*	0.02	0.058	0.02	0.151
16	3.08	4.24	5.99	3.08	*0.004*	3.08	0.032	3.08	0.257
17	6.80	4.92	0.92	6.8	*0.001*	4.92	0.048	0.92	0.698
18	3.97	0.56	0.59	3.97	*0.005*	0.56	0.054	0.56	0.369
19	58.02	19.22	57.16	58.02	*0.002*	19.22	0.021	19.22	0.145
20	21.89	5.62	39.26	21.89	*0.004*	5.62	0.037	5.62	0.079

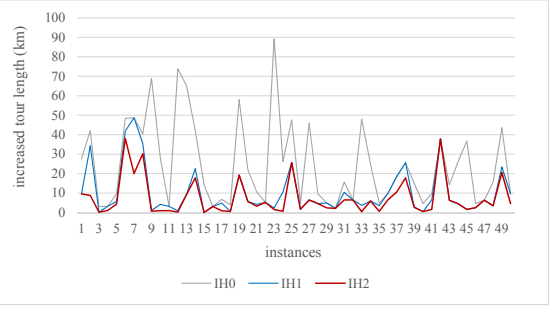


Fig. A. An increased tour length of IH0, IH1 and IH2 under fifty tested instances.

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